

CASCADING STYLE SHEET:

Program 1

Create a web page to show all hyperlinks with following specification: Default color is pink.

- ☐ Active color is blue.
- ☐ Visited color is Green.
- ☐ Hyperlink should be without underline.

Source CODE:

```
<!DOCTYPE html>

<html lang="en">

<head>

<meta charset="UTF-8">

<title>Hyperlink Styling</title>

<style>

a:link {

color: pink;

text-decoration: none;

}

a:visited {

color: green;

text-decoration: none;

}

a:active {

color: blue;

text-decoration: none;

}

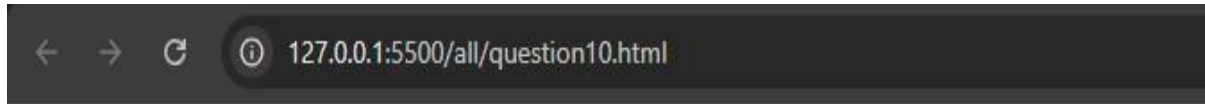
a:hover {
```

```
color: red;
text-decoration: none;

}

</style>
</head>
<body>
<h2>Styled Hyperlinks</h2>
<p>
  <a href="https://www.google.com" target="_blank">Google</a><br><br>
  <a href="https://www.wikipedia.org" target="_blank">Wikipedia</a><br><br>
  <a href="https://www.github.com" target="_blank">GitHub</a>
</p>
</body>
</html>
```

OUTPUT:



Styled Hyperlinks

[Google](#)

[Wikipedia](#)

[GitHub](#)

Program 2

Create Box Shadow and text Shadow using CSS3.

Source Code:

```
<!DOCTYPE html>

<html lang="en">

<head><meta charset="UTF-8">

<title>Box Shadow and Text Shadow</title>

<style>

.box {

width: 300px;

height: 150px;

background-color: rgb(236, 235, 144);

margin: 40px auto;

padding: 20px;

text-align: center;

box-shadow: 5px 5px 10px gray;

}

h2 {

text-align: center;

color: rgb(21, 173, 89);

text-shadow: 2px 2px 5px gray;}

</style>

</head>

<body><h2>This is the example of Text Shadow </h2>

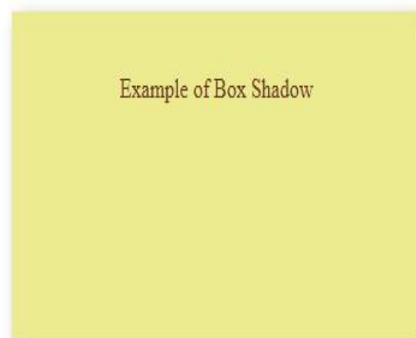
<div class="box">

<p style="color:rgba(71, 2, 2, 0.774)">Example of Box Shadow</p></div></body></html>
```

OUTPUT:



This is the example of Text Shadow



Program 3.

Create Rounded Corners using css3.

Source Code:

```
<!DOCTYPE html>
<html>
<head>
<title>Rounded Image</title>
<style>
.box {
width: 300px;

margin: 50px auto;

padding: 15px;

background-color: #f2f2f2;

text-align: center;

border-radius: 20px;

}
.box img {

width: 100%;

border-radius: 15px;

}

</style>
</head>
<body>
<div class="box">
<p style="color:rgba(95, 2, 2, 0.623)">Rounded Image using CSS3</p>
</div>

</body>

</html>
```

OUTPUT:



Program 4

Create a web page to show newspaper layout effects on contents given in web page (i.e. in multiple columns).

Source Code:

```
<!DOCTYPE html>

<html lang="en">

<head>

  <meta charset="UTF-8">

  <meta name="viewport" content="width=device-width, initial-scale=1.0">

  <title>Document</title>

  <style>

    h1,h3{

      text-shadow: 2px 10px 4px ;

    }

    .first{

      column-count: 3;

      display: flex;

    }

    .second{

      background-color: aqua;

      box-shadow: 10px 10px 10px ;

      text-align: left;

      border-radius: 10px;

    }

    .second{

      transition: all;

    }

    .second:hover{
```



```

        background-color: blueviolet;
        opacity: 2;
    }

    .box {
        width: 250px;
        padding: 25px;
        background-color: lightblue;
        text-align: center;
        font-size: 22px;
        cursor: pointer;
        border-radius: 10px;
        transition: transform 0.3s, box-shadow 0.3s;
    }

    .box:hover {
        transform: scale(1.1);
        box-shadow: 0 10px 20px rgba(0,0,0,0.3);
    }
</style>
</head>
<body>
    <h1>Toofan Express </h1>
    <div class="first">
        <div class="second">
            <h3>Dehradun</h3>

            Lorem ipsum dolor sit amet consectetur adipisicing elit. Voluptatibus cumque
            dolorum aliquid aperiam! Neque amet ducimus similique porro et soluta necessitatibus, ad
            eius eaque omnis repellat, magni possimus suscipit unde, nam beatae eos nihil laborum sed
            sunt. Tempora, vero exercitationem?

        </div>

```

```
<div class="second">
```

```
<h3>Dehli</h3>
```

Lorem ipsum dolor sit amet consectetur adipisicing elit. In doloribus magni eaque cupiditate odit modi sunt voluptatem culpa expedita laboriosam deleniti dolorem fuga at architecto minima, nisi debitis. Vero, eum! Labore corporis alias, earum fugiat laudantium est ipsum aut aperiam?

```
</div>
```

```
<div class="second">
```

```
<h3>Miyawala</h3>
```

Lorem ipsum dolor sit amet consectetur adipisicing elit. In doloribus magni eaque cupiditate odit modi sunt voluptatem culpa expedita laboriosam deleniti dolorem fuga at architecto minima, nisi debitis. Vero, eum! Labore corporis alias, earum fugiat laudantium est ipsum aut aperiam?

```
</div>
```

```
</div>
```

```
<br>
```

```
<!-- <br>
```

```
<h3>Pehle modi hai fir amit shah aayega fir yogi fir hemant vishwa sharma 2050 tk kahi ni  
jaara modi.</h3> -->
```

```
<hr style="height: 3px; color: black;">
```

```
<div class="second">
```

Lorem, ipsum dolor sit amet consectetur adipisicing elit. Et excepturi ipsam reprehenderit tempore quidem suscipit quod nemo officiis consequatur fuga amet ut accusantium, rerum, quam repudiandae dolores ullam unde quasi? Itaque alias fugiat recusandae quae nobis nostrum. Beatae, architecto illum.

```
</div>
```

```
<br><br>
```

```
</body>
```

```
</html>
```

OUTPUT:



Program 5:

Create a web page to show transition effect in such a way so that elements gradually change from one style to another style.

Source Code:

```
<!DOCTYPE html>

<html>

<head>

<title>CSS3 Transition Effect</title>

<style>

body {

font-family: Arial;

background-color: #f2f2f2;

padding: 30px;

}

h2 {

text-align: center;

}

.box {

width: 400px;

margin: 40px auto;

padding: 20px;

background-color: lightblue;

color: black;

text-align: justify;

transition:
```

```

background-color 1.5s,
color 1.5s,

transform 1.5s,

box-shadow 1.5s,

border-radius 1.5s;

}

.box:hover {

background-image: url("https://s3.amazonaws.com/images.seroundtable.com/google-amp-1454071566.jpg"); background-color: #ffb6c1;

color: rgb(239, 239, 245);

transform: scale(1.05);

box-shadow: 10px 10px 20px gray;

border-radius: 25px;

}

</style>

```

```
</head>
```

```
<body>
```

```
<h2>CSS3 Transition Example</h2>
```

```

<div class="box">CSS3 transition property allows elements to change from one style to another
gradually instead of changing suddenly. When the mouse pointer is placed over this box, you can
observe smooth changes in background color, text color, size, shadow, and rounded corners.
<br><br>

```

Transitions make web pages more interactive and user-friendly.

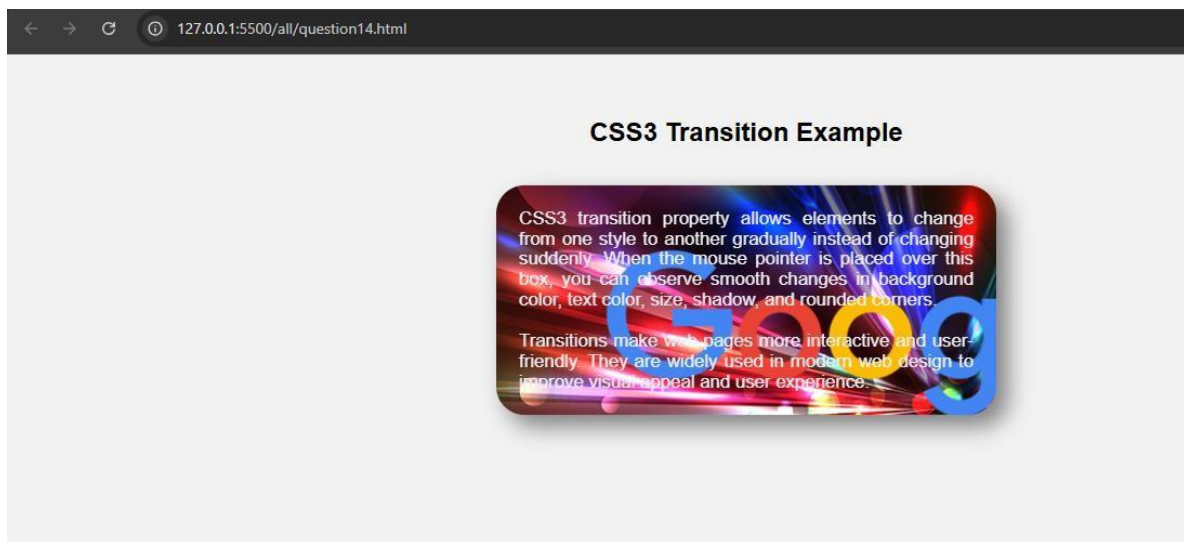
They are widely used in modern web design to improve visual appeal and user experience.

```

</div>
</body>
</html>

```

OUTPUT:



Program 6:

Create a web page to show fixed background image (this image will not scroll with the rest of the page).

Source Code:

```
<!DOCTYPE html>

<html>

<head>

<title>Fixed Image</title>

<style>

body{

    margin:0;

    font-family:Arial, sans-serif;

    background-image:url("par.jpg");

    background-size:cover;

    background-position:center;

    background-attachment:fixed;

    background-repeat:no-repeat;

}

.content{

    height:2000px;

    background:rgb(255,255,255,0.4);

    padding:20px;

}

</style>

</head>

<body>

<div class="content">

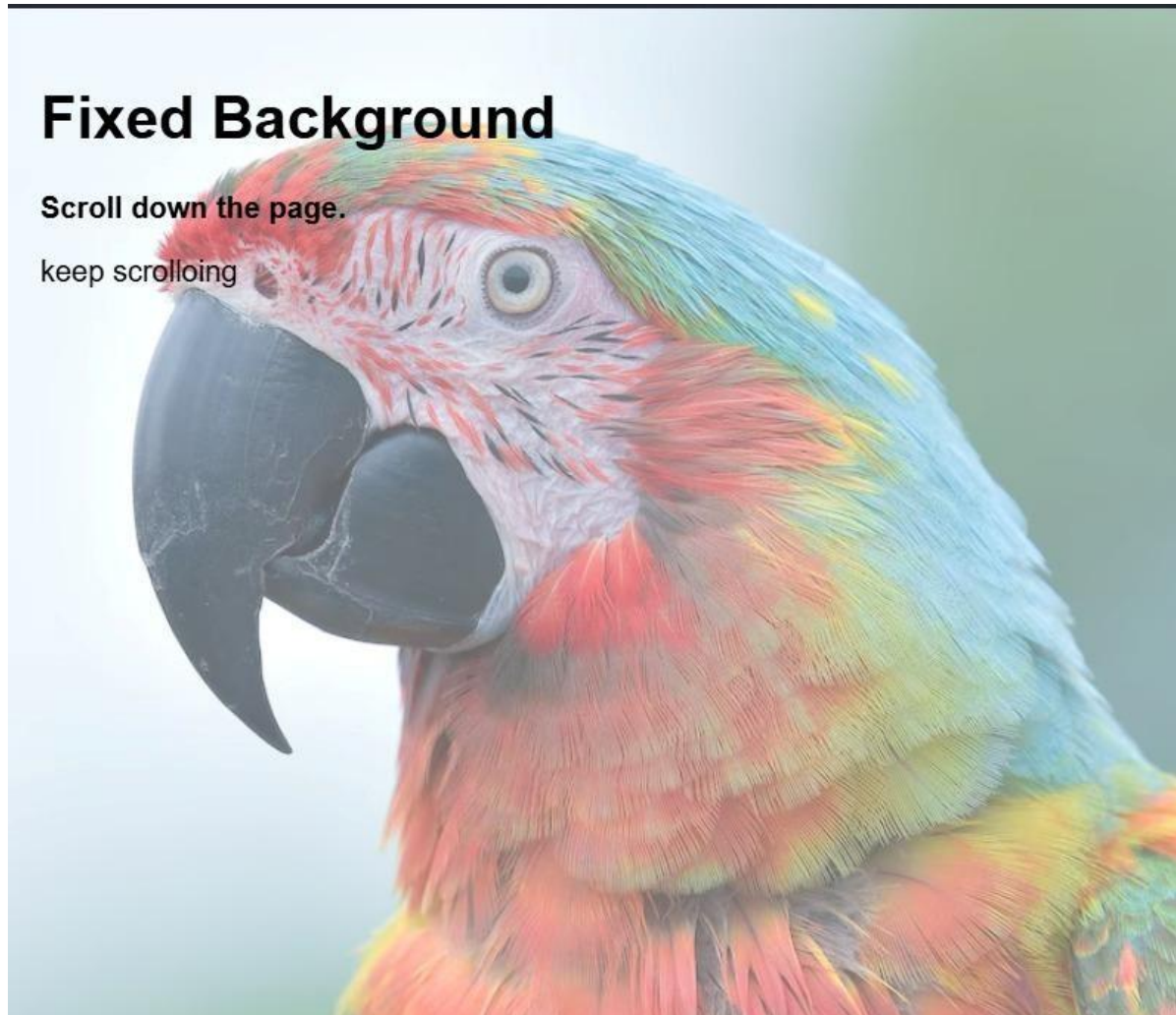
    <h1>Fixed Background </h1>

    <p> <b>Scroll down the page.</b></p>

    <p>keep scrolling</p>

</div></body></html>
```

OUTPUT:



Program 7:

Create a web page to position a background image and repeat the image horizontally or vertically.

Source Code:

```
<!DOCTYPE html>

<html>

<head>

<title>Background Image Repeat</title>

<style>

body{

    font-family:Arial;

}

.horizontal{

    height:200px;

    border:2px solid black;

    margin:20px;

    background-image:url("par.jpg");

    background-repeat:repeat-x;

    background-position:top left;

}

.vertical{

    height:300px;

    border:2px solid black;

    margin:20px;

    background-image:url("par.jpg");

    background-repeat:repeat-y;

    background-position:top left;

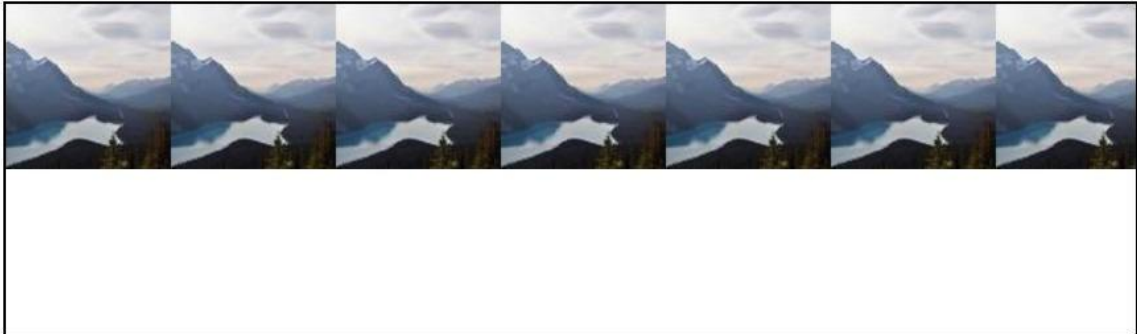
}

</style>
```

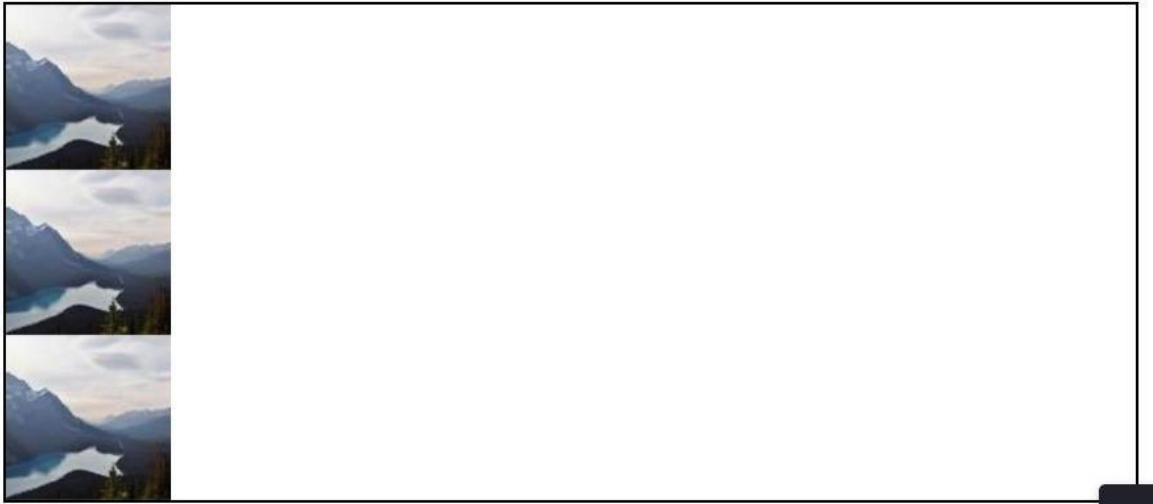
```
</head>
<body>
<h2>Horizontal Repeat</h2>
<div class="horizontal"></div>
<h2>Vertical Repeat</h2>
<div class="vertical"></div>
</body>
</html>
```

OUTPUT:

Horizontal Repeat



Vertical Repeat



Program 8:

Create an HTML page to demonstrate all types of CSS position properties (static, relative, absolute, fixed, and sticky).

Source Code:

```
<!DOCTYPE html>

<html>

<head>

<title>CSS Positions</title>

<style>

body{

    font-family:Arial;

    height:2000px;

    margin:0;

}

.box{

    width:200px;

    height:80px;

    color:white;

    padding:10px;

    margin:20px;

}

.static{

    position:static;

    background:gray;

}

.relative{

    position:relative;

    left:40px;

    top:10px;

    background:green;
```

```
}  
.container{  
    position:relative;  
    height:200px;  
    border:2px solid black;  
    margin:20px;  
}  
.absolute{  
    position:absolute;  
    right:10px;  
    top:10px;  
    background:red;  
}  
.fixed{  
    position:fixed;  
    bottom:10px;  
    right:10px;  
    background:blue;  
}  
.sticky{  
    position:sticky;  
    top:0;  
    background:orange;  
}  
</style>  
</head>
```

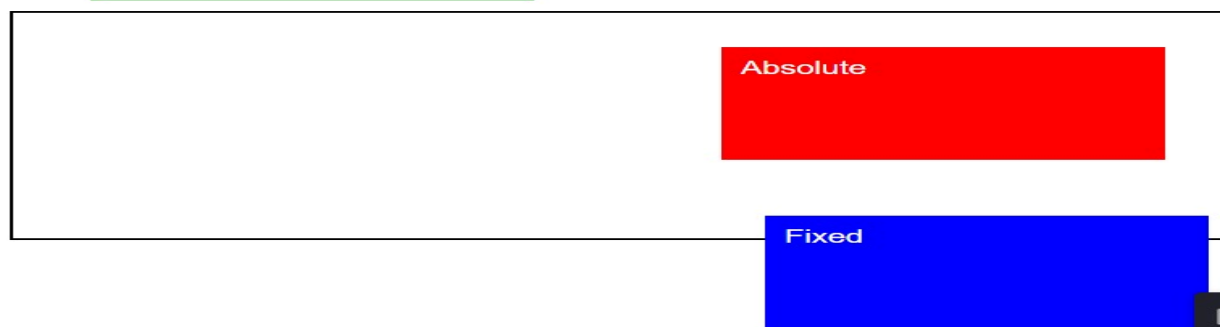
```
<body>
<h2 class="box sticky">Sticky</h2>
<div class="box static">
  Static
</div>
<div class="box relative">
  Relative
</div>
<div class="container">
  <div class="box absolute">
    Absolute
  </div>
</div>
<div class="box fixed">
  Fixed
</div>
</body>
</html>
```

OUTPUT:

Sticky

Static

Relative



Fixed

Program 9:

Design a web page using CSS which includes the following:

- i. Use different font styles
- ii. Set background image for both the page and single elements on page.
- iii. Control the repetition of image with background-repeat property
- iv. Define style for links as a:link, a:active, a:hover, a:visited
- v. Add customized cursors for links.
- vi. Work with layers.

Source Code:

```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<title>Web Page using CSS</title>
<style>
body{
background-image: url("https://picsum.photos/1200/800");
background-repeat: no-repeat;
background-size: cover;
font-family: Arial, sans-serif;
}
h1{
font-family: Georgia, serif; text-
align: center;
color: darkblue;
}
p{
font-family: "Courier New", monospace;
font-size: 16px;
}
```



```
.box {
background-image: url("https://picsum.photos/400/200");
background-repeat: repeat-x;
padding: 15px;
margin: 20px; color:
white;
}
a:link {
color: blue;
text-decoration: none;
}
a:visited {
color:purple;
}
a:hover { color:
red;
text-decoration: underline;
cursor: url("https://cur.cursors-4u.net/cursors/cur-2/cur117.cur"), pointer;
}
a:active {
color:green;
}
.layer1 {
position: absolute;
top:300px;
left: 200px;
font-size: 40px; color:
blue;
z-index: 1;
}
.layer2 {
```

```

position: absolute; top:
320px;
left: 220px; font-size:
40px; color: red;
z-index: 2;
}
</style>
</head>
<body>
<h1>Web Page using CSS</h1>
<div class="box">

<p>Welcome to Web page using CSS.CSS stands for Cascading Style Sheets and is used
to style web pages.</p>

</div>
<center>

    <a href="#">Register</a>

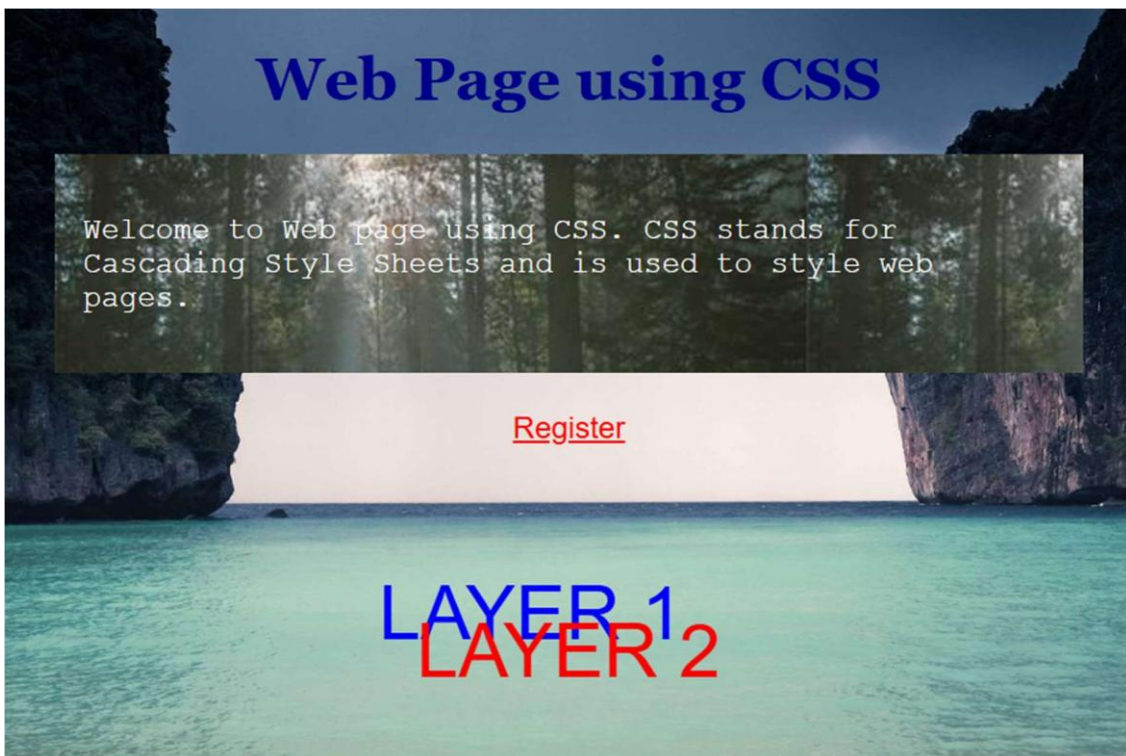
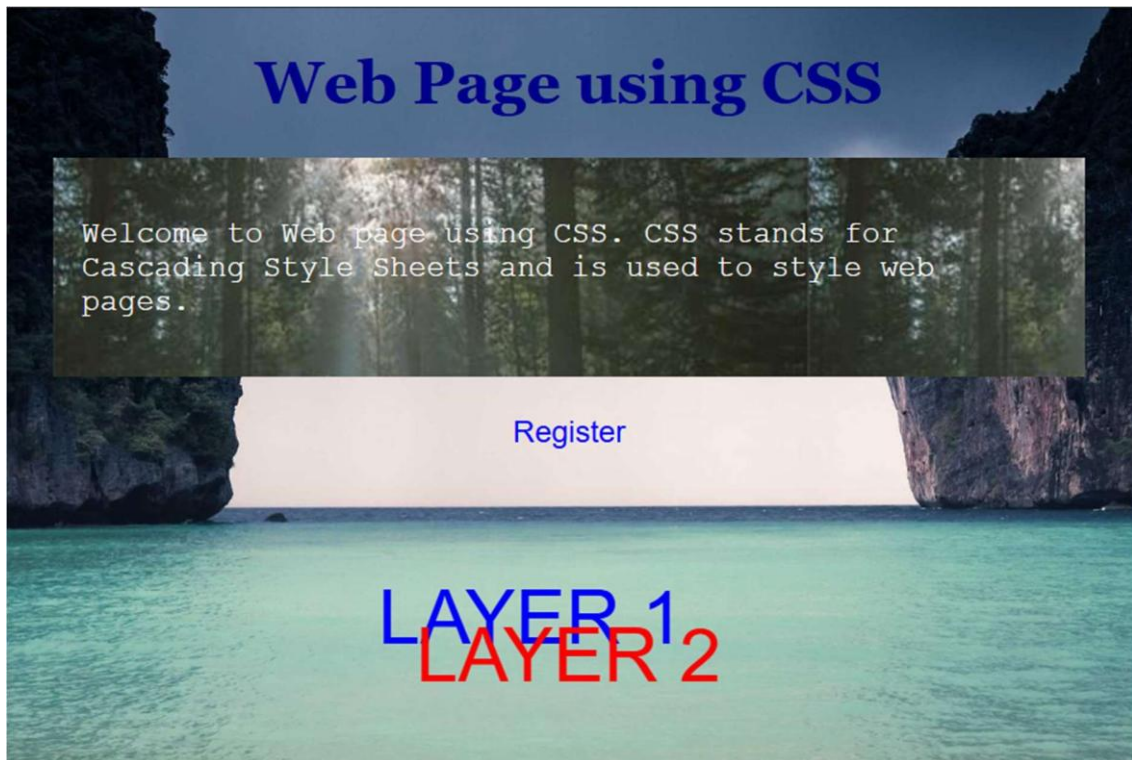
</center>

<div class="layer1">LAYER 1</div>
<div class="layer2">LAYER 2</div>

</body>
</html>

```

OUTPUT



JavaScript

Q1: Design a web page to validate credit card numbers per the specifications below. The following tables outline the major credit cards you want to validate and allow prefixes and lengths.

Card Type	Prefix	Length
Master Card	51–55	16
Visa	4	13,16
American Express	34,37	15

Index.html

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Credit Card Validation</title>
  <script src="script.js"></script>
</head>
<body>
  <div>
    <fieldset>
      <legend>Credit Card Validation</legend>
      <label>Card Type:</label>
      <input type="text" id="cardt" placeholder="Visa / MasterCard / American Express">
      <br><br>
      <label>Card Number:</label>
      <input type="text" id="cardn">
      <br><br>
      <button onclick="validateCard()">Validate</button>
      <br><br>
      <div id="output"></div>
```

```
        </fieldset>
    </div>
</body>
</html>
```

Script.js

```
function validateCard() {
    let cardType = document.getElementById("cardt").value.toLowerCase();
    let cardNumber = document.getElementById("cardn").value;
    let len = cardNumber.length;
    let result = "";
    if (cardType == "visa") {
        if (cardNumber.startsWith("4") && (len === 13 || len === 16)) {
            result = "Valid Visa Card";
        } else {
            result = "Invalid Visa Card";
        }
    }
    else if (cardType === "mastercard") {
        let prefix = parseInt(cardNumber.substring(0, 2));
        if (len === 16 && prefix >= 51 && prefix <= 55) {
            result = "Valid MasterCard";
        } else {
            result = "Invalid MasterCard";
        }
    }
    else if (cardType === "american express") {
        if (len === 15 && (cardNumber.startsWith("34") || cardNumber.startsWith("37"))) {
            result = "Valid American Express Card";
        } else {
```

```

        result = "Invalid American Express Card";
    }
}
else {
    result = "Unknown Card Type";
}
document.getElementById("output").innerHTML = result;
}

```

OUTPUT:



Q 2: Design a web page to validate the following according to the standard conditions.

a. Name

b. E-Mail-id

c. Password.

Index.html:

```
<!DOCTYPE html>

<html lang="en">

<head>

    <meta charset="UTF-8">

    <meta name="viewport" content="width=device-width, initial-scale=1.0">

    <title>Document</title>

    <script src="script.js"></script>

</head>

<body>

    <fieldset method="get">

        <legend>MAIL VERIFIER</legend>

        <label for="Name">NAME:</label>

        <input type="text" id="name" placeholder="Enter your name"><br> <br>

        <label for="E-mail">Email:</label>

        <input type="text" id="email" placeholder="Enter your mail"><br><br>

        <label for="Password">Password:</label>

        <input type="password" id="password" placeholder="Enter password"> <br><br>

        <button onclick="x()">Verify</button><br><br>

        <div id="output"></div>

    </fieldset>

</body>

</html>
```

Script.js

```
function x(){
```

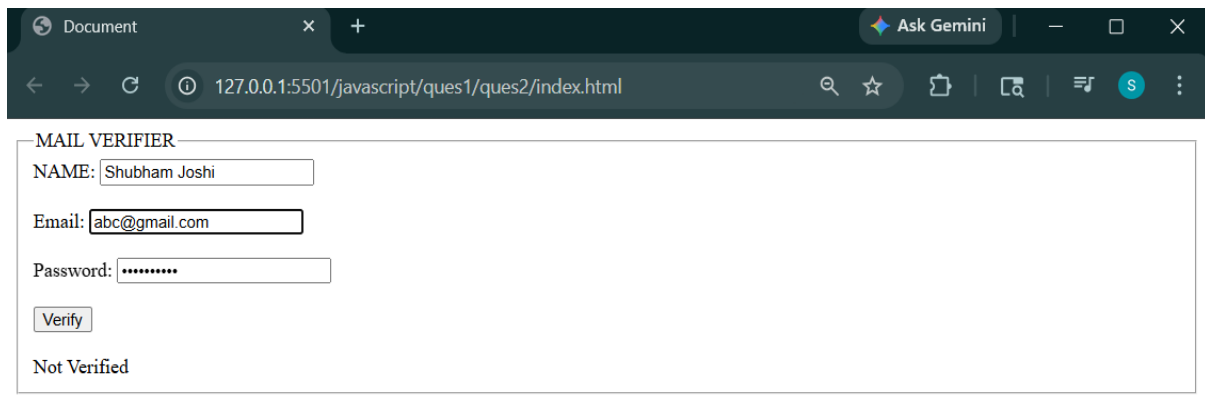
```

let e,p,result;

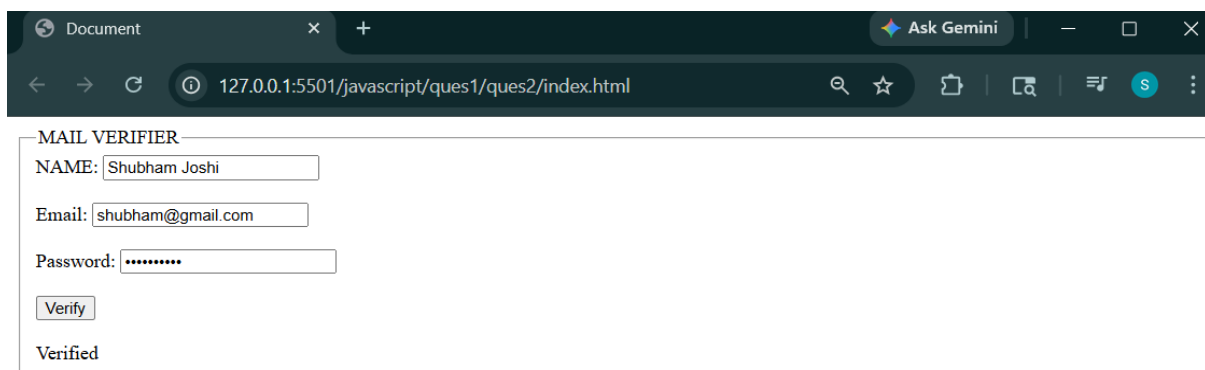
e=document.getElementById("email").value.toLowerCase();
p=document.getElementById("password").value;
if(!e.endsWith("@gmail.com")){
    result="Invalid Email";
}
else{
    if(e=="shubham@gmail.com"&&p=="shubham@01")
        result="Verified";
    else if(e=="rohit@gmail.com"&&p=="rohit@01")
        result="Verified";
    else
        result="Not Verified";
}
document.getElementById("output").innerHTML=result;
}

```

OUTPUT:



A screenshot of a web browser window titled 'Document'. The address bar shows '127.0.0.1:5501/javascript/ques1/ques2/index.html'. The page content is a form titled 'MAIL VERIFIER' with the following fields: 'NAME: Shubham Joshi', 'Email: abc@gmail.com', and 'Password:'. Below the fields is a 'Verify' button. The output below the button is 'Not Verified'.



A screenshot of a web browser window titled 'Document'. The address bar shows '127.0.0.1:5501/javascript/ques1/ques2/index.html'. The page content is a form titled 'MAIL VERIFIER' with the following fields: 'NAME: Shubham Joshi', 'Email: shubham@gmail.com', and 'Password:'. Below the fields is a 'Verify' button. The output below the button is 'Verified'.

Q3: Store some country names and their capitals. Ask the user to select a country and its capital from two lists. If the match is correct, display “Correct answer”; otherwise, display an error message and tell the correct answer.

Index.html:

```
<!DOCTYPE html>

<html lang="en">

<head>

  <meta charset="UTF-8">

  <meta name="viewport" content="width=device-width, initial-scale=1.0">

  <title>Country Capital Quiz</title>

  <style>

    body {

      font-family: Arial, sans-serif;

      text-align: center;

      margin-top: 50px;

    }

    select, button {

      padding: 10px;

      margin: 10px;

      font-size: 16px;

    }

    #output {

      margin-top: 20px;

      font-size: 20px;

      font-weight: bold;

    }

  </style>

  <script src="script.js"></script>

</head>

<body>
```

```

<h2>Country & Capital Quiz</h2>
<label>Country:</label><br>
<select id="country">
  <option value="">-- Select Country --</option>
  <option value="India">India</option>
  <option value="Russia">Russia</option>
  <option value="Japan">Japan</option>
  <option value="America">America</option>
  <option value="China">China</option>
</select><br>
<label>Capital:</label><br>
<select id="capital">
  <option value="">-- Select Capital --</option>
  <option value="New Delhi">New Delhi</option>
  <option value="Moscow">Moscow</option>
  <option value="Tokyo">Tokyo</option>
  <option value="Washington D.C">Washington D.C</option>
  <option value="Beijing">Beijing</option>
</select> <br>
<button onclick="verify()">Verify</button>
<div id="output"></div>
</body>
</html>

```

Script.js

```

function verify() {
  let c = document.getElementById("country").value;
  let cap = document.getElementById("capital").value;
  let output = document.getElementById("output");
  const capitals = {
    "India": "New Delhi",

```

```

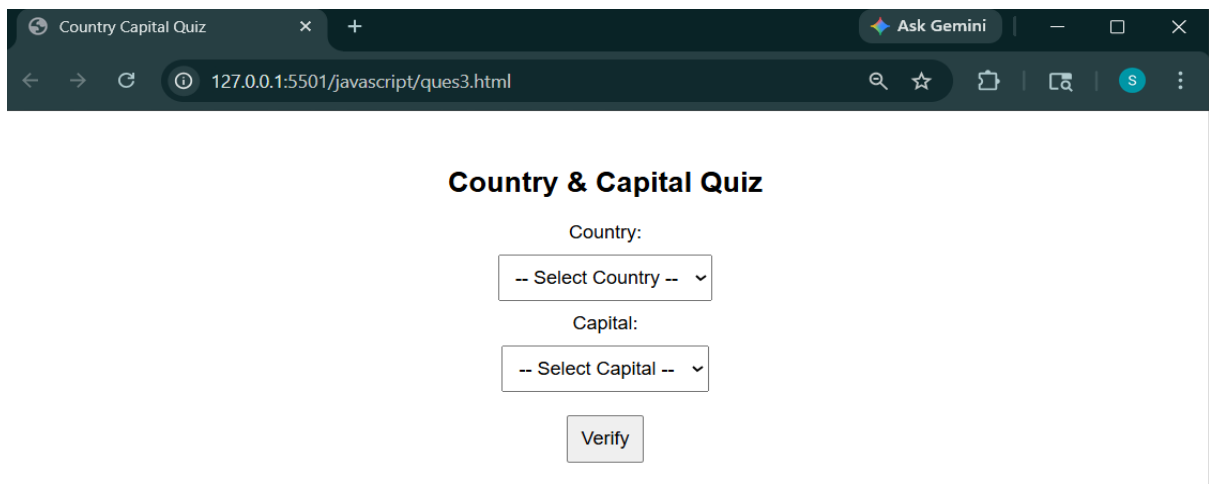
    "America": "Washington D.C",
    "Japan": "Tokyo",
    "Russia": "Moscow",
    "China": "Beijing"
};

if (c === "" || cap === "") {
    output.innerHTML = "Please select both country and capital";
    output.style.color = "orange";
    return;
}

if (capitals[c] === cap) {
    output.innerHTML = "Correct Answer!";
    output.style.color = "green";
} else {
    output.innerHTML = "Wrong Answer!";
    output.style.color = "red";
}
}

```

OUTPUT:



Country Capital Quiz

Country:

-- Select Country --

Capital:

-- Select Capital --

Verify

Country & Capital Quiz

Country:

Japan

Capital:

Tokyo

Verify

Correct Answer!

Country & Capital Quiz

Country:

India

Capital:

Beijing

Verify

Wrong Answer!

Q 4: Design the simple Calculator.

Index.html:

```
<!DOCTYPE html>

<html lang="en">

<head>

  <meta charset="UTF-8">

  <meta name="viewport" content="width=device-width, initial-scale=1.0">

  <title>Calculator</title>

  <style>

    body{

      display: flex;

      justify-content: center;

      align-items: center;

      height: 100vh;

      background: #f2f2f2;

      font-family: Arial, sans-serif;

    }

    .calculator{

      background: #ffffff;

      padding: 15px;

      border-radius: 10px;

      box-shadow: 0 8px 20px rgba(0,0,0,0.15);

    }

    #show{

      width: 100%;

      height: 45px;

      font-size: 18px;

      text-align: right;

      margin-bottom: 10px;

      padding-right: 8px;
```

```

    }
    button{
        width: 55px;
        height: 45px;
        margin: 3px;
        font-size: 16px;
        border: none;
        border-radius: 6px;
        cursor: pointer;
        background: #e0e0e0;
    }
</style>

<script src="script.js"></script>
</head>
<body>
<div class="calculator">
    <input type="text" id="show">
    <br>
    <button onclick="x('(')"></button>
    <button onclick="x(')')"></button>
    <button onclick="x('%')">%</button>
    <button onclick="del()">&cross;</button>
    <button class="clear" onclick="document.getElementById('show').value="">AC</button>
    <br>
    <button onclick="x('1')">1</button>
    <button onclick="x('2')">2</button>
    <button onclick="x('3')">3</button>
    <button class="op" onclick="x('+')">+</button>
    <br>
    <button onclick="x('4')">4</button>

```

```

<button onclick="x('5')">5</button>
<button onclick="x('6')">6</button>
<button class="op" onclick="x('-)">-</button>
<br>
<button onclick="x('7')">7</button>
<button onclick="x('8')">8</button>
<button onclick="x('9')">9</button>
<button class="op" onclick="x('*')">*</button>
<br>
<button onclick="x('0')">0</button>
<button onclick="x('.')">.</button>
<button class="equal" onclick="calculate()">=</button>
<button class="op" onclick="x('/')">/</button>
</div>
</body>
</html>

```

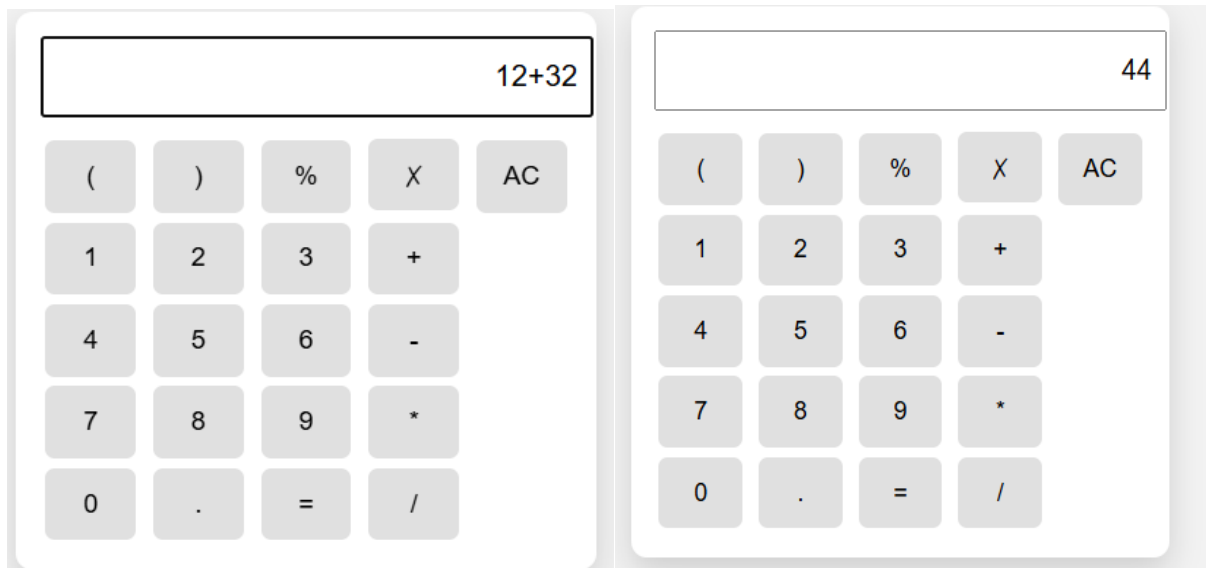
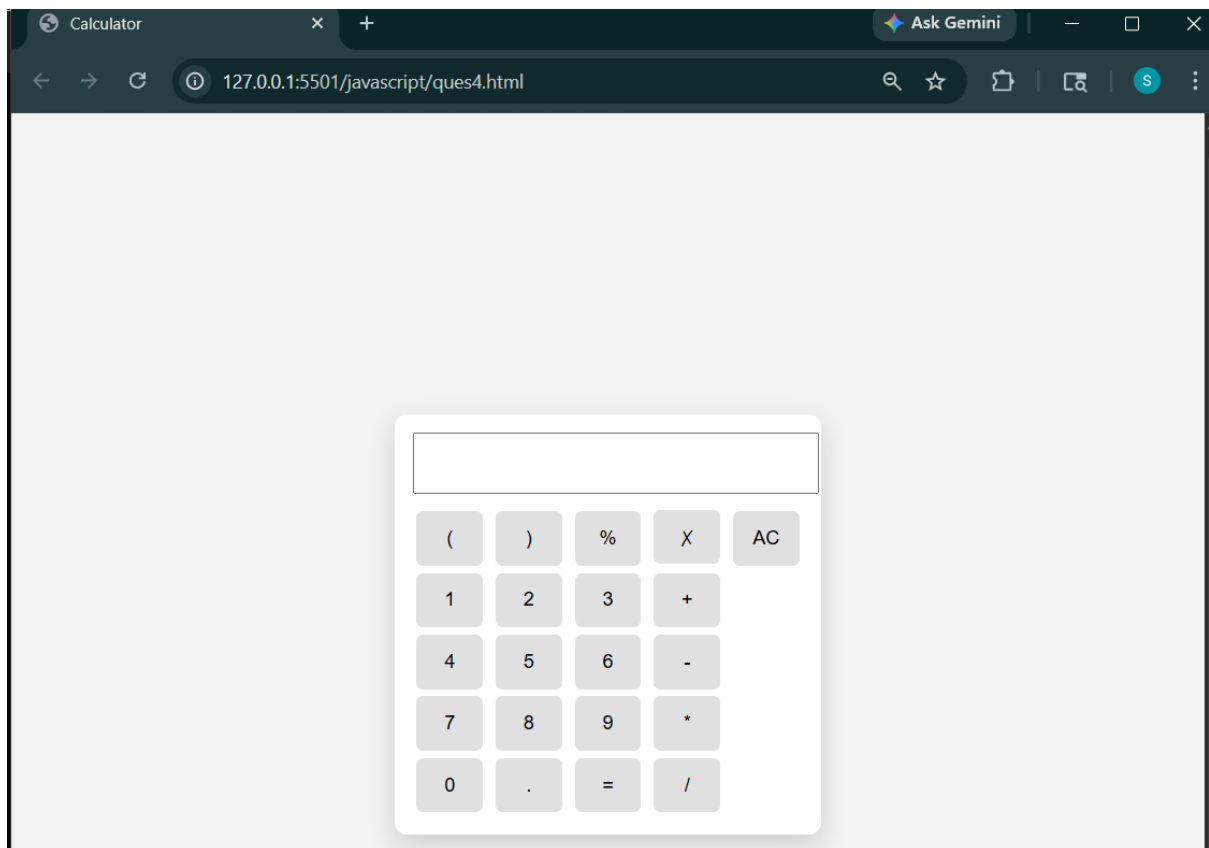
Script.js

```

function x(a){
    document.getElementById("show").value += a;
}
function calculate(){
    let result = eval(document.getElementById("show").value);
    document.getElementById("show").value = result;
}
function del(){
    let box = document.getElementById("show");
    box.value = box.value.slice(0, -1);
}

```

OUTPUT:



Index.html:

Himanshu Raturi

```
</style>

<script src="script.js"></script>

<link rel="stylesheet" href="style.css">

</head>

<body>

  <div id="clk" class="time">

  </div>

</body>

</html>
```

Script.js

```
function x(){

  let x = "#" + Math.floor(Math.random() * 123456).toString(16).padStart(6, '0');

  document.getElementById("na").style.backgroundColor=x;

}

function y(){

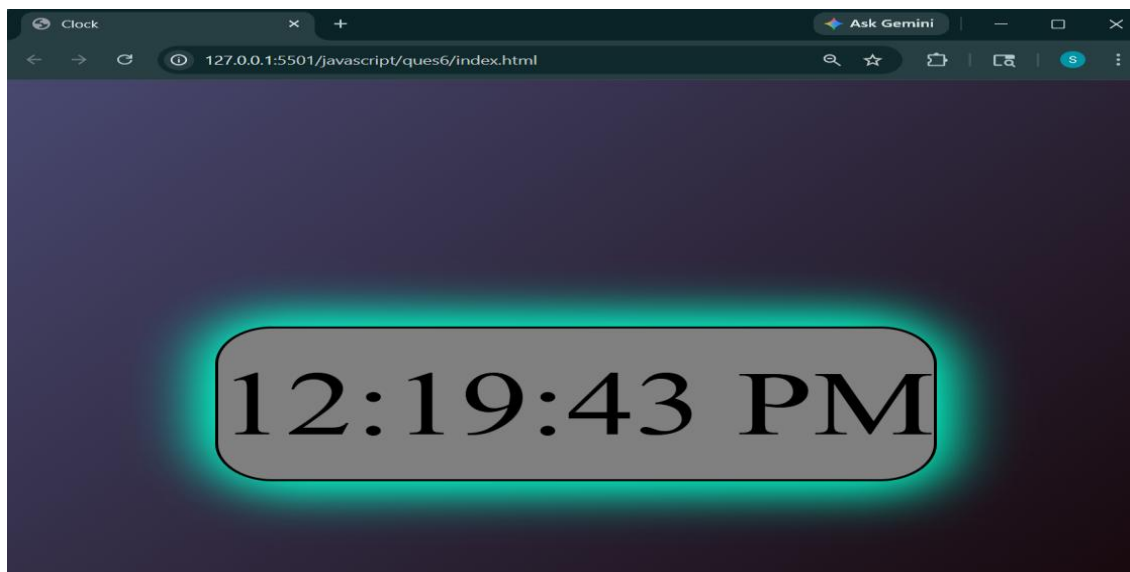
  let now= new Date().toLocaleTimeString();

  document.getElementById("clk").innerHTML=now;

}

setInterval(y,1000);
```

OUTPUT:



Q 7: Write a code for a web application that accepts the user's birthdate in a textbox and displays the day of the week in a message box at the click of a button

Index.html:

```
<!DOCTYPE html>

<html lang="en">

<head>

  <meta charset="UTF-8">

  <meta name="viewport" content="width=device-width, initial-scale=1.0">

  <title>Birthday Day Finder</title>

  <style>

    body {

      font-family: Arial, sans-serif;

      text-align: center;

      margin-top: 50px;

    }

    .container {

      display: inline-block;

    }

    input, button {

      padding: 10px;

      font-size: 16px;

    }

    .modal {

      display: none;

      position: fixed;

      top: 0;

      left: 0;

      width: 100%;

      height: 100%;

      background: rgba(0,0,0,0.5);

    }

  </style>


```

```

.modal-content {
    background: white;
    padding: 20px;
    margin: 15% auto;
    width: 300px;
    border-radius: 10px;
}
</style>
<script src="script.js"></script>
</head>

<body>
    <div class="container">
        <fieldset>
            <legend>Info</legend>
            <label>Birthday:</label>
            <input type="text" id="bd" placeholder="YYYY-MM-DD">
            <br><br>
            <button onclick="findDay()">Find Day</button>
        </fieldset>
    </div>
    <div id="modal" class="modal">
        <div class="modal-content">
            <p id="message"></p>
            <button onclick="closeModal()">OK</button>
        </div>
    </div>
</body>
</html>

```

Script.js

```

function findDay() {
    let x = document.getElementById("bd").value;
    let message = document.getElementById("message");
    let modal = document.getElementById("modal");
    let dt = new Date(x);

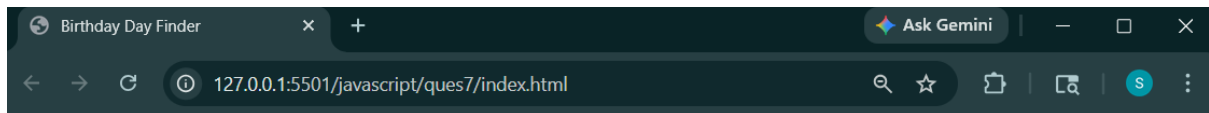
    let days = ["Sunday", "Monday", "Tuesday", "Wednesday", "Thursday", "Friday",
    "Saturday"];

    if (!x || isNaN(dt.getTime())) {
        message.innerHTML = "Invalid date! Use format YYYY-MM-DD";
    } else {
        let day = days[dt.getDay()];
        message.innerHTML = "You were born on: <b>" + day + "</b>";
    }
    modal.style.display = "block";
}

function closeModal() {
    document.getElementById("modal").style.display = "none";
}

```

OUTPUT:



Info

Birthday:



Q 8: Write a script that inputs a telephone number as a string in the form (555)555-555. The script should use the strings method split to extract the area code as a token and the last four digits of the phone numbers as a token. Display the area code in one test field and the seven-digit phone number in another text field.

Index.html:

```
<!DOCTYPE html>

<html lang="en">

<head>

  <meta charset="UTF-8">

  <meta name="viewport" content="width=device-width, initial-scale=1.0">

  <title>Mobile Number Verifier</title>

  <style>

    body {

      font-family: Arial;

      text-align: center;

      margin-top: 50px;

    }

    input {

      padding: 8px;

      font-size: 16px;

    }

    #output {

      margin-top: 20px;

    }

  </style>

  <script src="script.js"></script>

</head>

<body>

  <div>

    <div id="input">

      <label for="Mob no">Mobile Number:</label><br><br>
```

```

        <input type="text" id="mob" placeholder="(555)555-555"><br><br>
        <input type="submit" onclick="verify()"><br><br>
    </div>

    <div id="output">
        <label for="" >Area Code: </label>
        <input type="text" name="" id="area-code"><br><br>
        <label for="">Remaining mobile no:</label>
        <input type="text" name="" id="rem-num">
    </div>
</div>
</body>
</html>

```

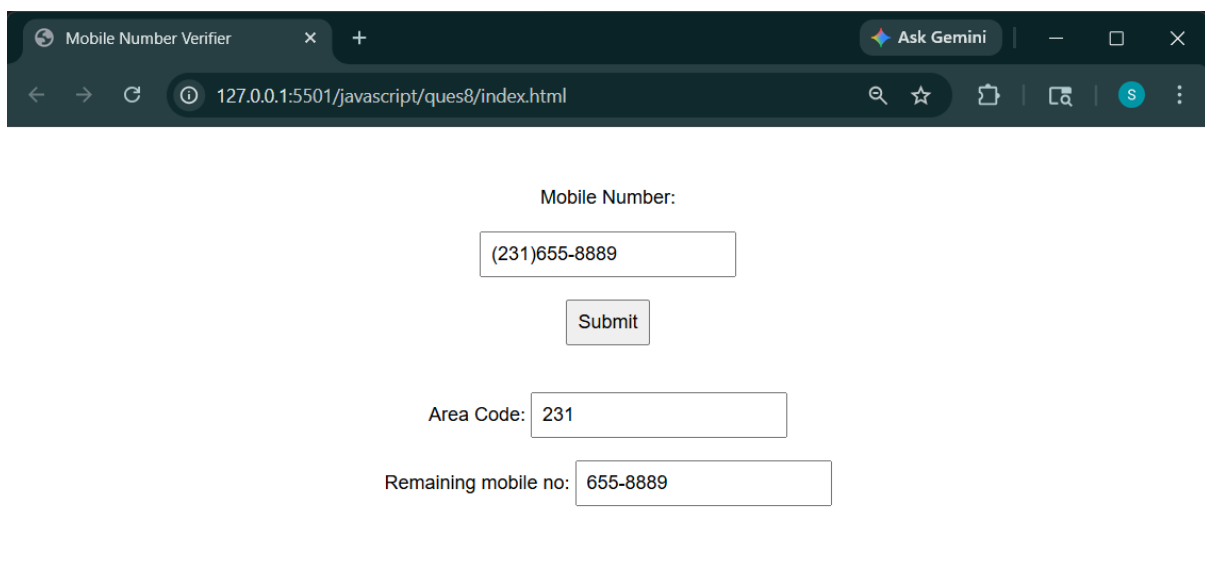
Script.js

```

function verify(){
    let n=document.getElementById("mob").value;
    let ac=n.split("(");
    document.getElementById("area-code").value=ac[0].split(" ")[1];
    document.getElementById("rem-num").value=ac[1];
}

```

OUTPUT:



Q 9: Develop and demonstrate an HTML file that includes a JavaScript script that uses functions for the following problems:

- a. Parameter: A string Output: The position in the string of the left-most vowel**
- b. Parameter: A number Output: The number with its digits in the reverse order.**

Index.html:

```
<!DOCTYPE html>

<html lang="en">

<head>

  <meta charset="UTF-8">

  <meta name="viewport" content="width=device-width, initial-scale=1.0">

  <title>Document</title>

  <script src="script.js"></script>

</head>

<body>

  <div style="border: solid 2px;">

    <h1>Question 9</h1>

    <div id="str" style="border: solid 2px; border-radius: 50px; text-align: center;"><br>

      <label for="string">Enter the String:</label>

      <input type="text" id="st"><br><br>

      <label for="Output">Position of leftmost vowel:</label>

      <input type="text" id="pos"><br><br>

      <button type="submit" onclick="findpos()">find</button>

      <br><br>

    </div>

    <div id="int" style="border: solid 2px; border-radius: 50px; text-align: center;"><br>

      <label for="string">Enter the Number:</label>

      <input type="text" id="num"><br><br>

      <label for="Output">Number in reverse order:</label>

      <input type="text" id="rev"><br><br>

      <button type="submit" onclick="findreverse()">find</button><br><br>

    </div>

  </body>

</html>
```

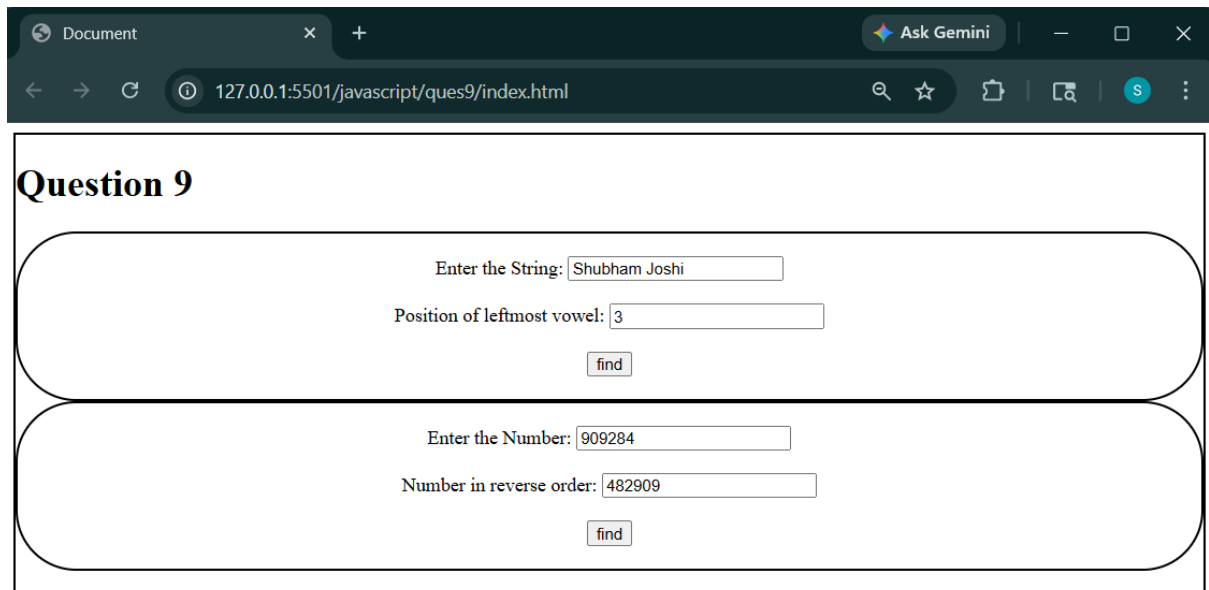
```
<br>
</div>
</div>
</body>
</html>
```

Script.js

```
function findpos(){
    let str1=document.getElementById("st").value;
    let n=str1.length;
    for(let i=0;i<n;i++){
        if("aeiouAEIOU".includes(str1[i])){
            document.getElementById("pos").value=i+1;
            return;
        }
    }
    document.getElementById("pos").value="vowel does not exist in the string";
}

function findreverse(){
    let str1=document.getElementById("num").value;
    let n=str1.length;
    let arr=str1.split("");
    for(let i=0;i<n/2;i++){
        let t=arr[i];
        arr[i]=arr[n-i];
        arr[n-i]=t;
    }
    str1=arr.join("");
    document.getElementById("rev").value=str1;
}
```

OUTPUT:



The screenshot shows a web browser window with a dark theme. The address bar displays '127.0.0.1:5501/javascript/ques9/index.html'. The page content is titled 'Question 9' and contains two distinct input sections, each enclosed in a rounded rectangle.

Section 1:

Enter the String:

Position of leftmost vowel:

Section 2:

Enter the Number:

Number in reverse order:

Q 10: Write a JavaScript function that takes a string that has lower and upper case letters as a parameter and converts upper case letters to lower case and lower case letters to upper case.

Index.html:

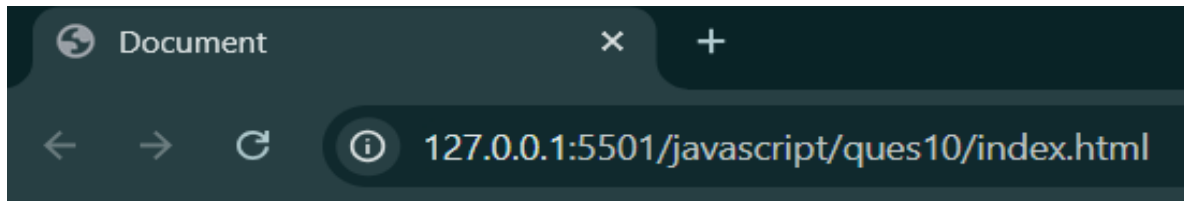
```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Document</title>
  <script src="script.js"></script>
</head>
<body>
  <label for="input">Enter the string:</label>
  <input type="text" id="in"><br><br>
  <button onclick="find()">reverse</button><br><br>
  <label for="output">String after operation:</label>
  <input type="text" id="ot">
</body>
</html>
```

Script.js

```
function find(){
  let s=document.getElementById("in").value;
  let n=s.length;
  let output="";
  for(let i=0;i<n;i++){
    if(s[i].toUpperCase()!=s[i]){
      output+=s[i].toUpperCase();
    }
    else{
```

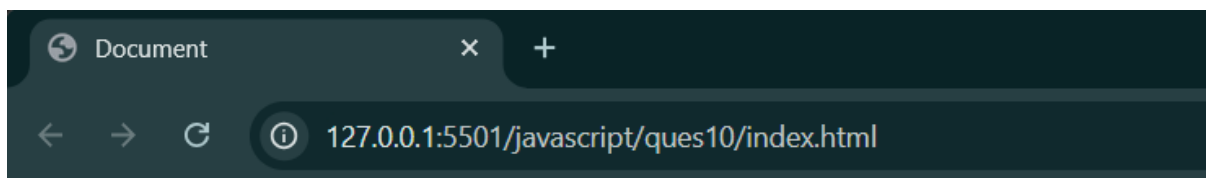
```
        output+=s[i].toLowerCase();
    }
}
document.getElementById("ot").value=output;
}
```

OUTPUT:



Enter the string:

String after operation:



Enter the string:

String after operation:

Q 11: Design a web page to perform a survey on four different models of Maruti (Maruti -K10, Zen-Astelo, Wagnor, Maruti- SX4) owned by people living in four metro cities(Delhi, Mumbai, Chennai & Kolkatta). Display tabulated report like format given below:

	Maruti-K10	Zen-Astelo	Wagnor	Maruti-SX4
Delhi				
Mumbai				
Cheennai				
Kolkatta				

Calculate the number of cars of different models in each metro city.

Index.html:

```
<!DOCTYPE html>
```

```
<html lang="en">
```

```
<head>
```

```
  <meta charset="UTF-8">
```

```
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
```

```
  <title>Document</title>
```

```
<style>
```

```
  body {
```

```
    margin: 0;
```

```
    font-family: Arial, sans-serif;
```

```
    text-align: center;
```

```
    background-color: #f5f5f5;
```

```
  }
```

```
  fieldset {
```

```
    width: 400px;
```

```
    margin: 40px auto;
```

```
    padding: 20px;
```

```
    border: 2px solid #444;
```

```
    border-radius: 10px;
```

```
    background-color: #fffaf0;
```

```
    box-shadow: 0 4px 10px rgba(0, 0, 0, 0.2);
}

legend {
    font-weight: bold;
    font-size: 18px;
    padding: 5px 10px;
    color: #333;
}

table {
    margin: 20px auto;
    background-color: #f0f0f0;
    border-collapse: collapse;
}

th, td {
    padding: 10px 20px;
    text-align: center;
    border: 1px solid #ccc;
}

.controls {
    margin-top: 20px;
}

select, button {
    padding: 8px 12px;
    margin: 5px;
    font-size: 14px;
}

button {
    cursor: pointer;
    background-color: #444;
    color: white;
```

```

        border: none;

        border-radius: 5px;
    }
</style>

<script src="script.js"></script>
</head>

<body>

    <div>

        <h1>Car Survey</h1>

        <table style="border: solid 2px;" border="2px">
            <tr>
                <th></th>
                <th>Maruti-K 10</th>
                <th>Zen-astelo</th>
                <th>Wagnor</th>
                <th>Maruti-SX4</th>
            </tr>
            <tr>
                <td>Dehli</td>
                <td id="dm"></td>
                <td id="dz"></td>
                <td id="dw"></td>
                <td id="dms"></td>
            </tr>
            <tr>
                <td>Mumbai</td>
                <td id="mm"></td>
                <td id="mz"></td>
                <td id="mw"></td>
                <td id="mms"></td>
            </tr>
        </table>
    </div>

```



```

</tr>
<tr>
  <td>Chennai</td>
  <td id="cm"></td>
  <td id="cz"></td>
  <td id="cw"></td>
  <td id="cms"></td>
</tr>
<tr>
  <td>Kolkata</td>
  <td id="km"></td>
  <td id="kz"></td>
  <td id="kw"></td>
  <td id="kms"></td>
</tr>
</table><br>
<fieldset>
  <legend>CarSurvey</legend>
  <b><label for="">Car:</label></b>
  <select name="" id="car">
    <option value="Maruti-K10">Maruti-k10</option>
    <option value="Zen-astelo">Zen-astelo</option>
    <option value="Wagnor">Wagnor</option>
    <option value="Maruti-SX4">Maruti-SX4</option>
  </select><br><br>
  <b><label for="">City:</label></b>
  <select name="" id="city">
    <option value="Delhi">Delhi</option>
    <option value="Mumbai">Mumbai</option>
    <option value="Chennai">Chennai</option>

```

```

        <option value="Kolkata">Kolkata</option>
    </select><br><br>
    <button onclick="add()">Modify</button>
</fieldset>
</div>
</body>
</html>

```

Script.js

```

function add(){
    let c={
        "Delhi":'d',
        "Mumbai":'m',
        "Chennai":'c',
        "Kolkata":'k'
    }
    let m={
        "Maruti-K10":'m',
        "Zen-astelo":'z',
        "Wagnor":'w',
        "Maruti-SX4":'ms'
    }
    let cc=document.getElementById("city").value;
    let mc=document.getElementById("car").value;
    let id="";
    id+=c[cc];
    id+=m[mc];
    let n=Number(document.getElementById(id).innerHTML);
    n+=1;
    document.getElementById(id).innerHTML=n;
    let sound = document.getElementById("engineSound");

```

```
sound.currentTime = 0;  
sound.play();  
let car = document.getElementById("carAnimation");  
car.classList.remove("move");  
void car.offsetWidth;  
car.classList.add("move");}
```

OUTPUT:

Document 127.0.0.1:5501/javascript/ques11/index.html

Car Survey

	Maruti-K 10	Zen-astelo	Wagnor	Maruti-SX4
Dehli				
Mumbai				
Chennai				
Kolkata				

CarSurvey

Car:

City:

Modify

Document 127.0.0.1:5501/javascript/ques11/index.html

Car Survey

	Maruti-K 10	Zen-astelo	Wagnor	Maruti-SX4
Dehli		1		
Mumbai			2	
Chennai		2		3
Kolkata	3			

CarSurvey

Car:

City:

Modify

PHP Program

Question: Create a web page to maintain a session using PHP.

```
<!DOCTYPE html>

<html>

<head>

    <title>Session Form</title>

</head>

<body>

    <h1>Session Management Form</h1>

    <form method="POST" action="session_handler.php">

        <label>Name:</label><br>

        <input type="text" name="username" required><br><br>

        <label>Email:</label><br>

        <input type="email" name="email" required><br><br>

        <button type="submit">Submit</button>

    </form>

</body>

</html>
```

Session_handler.php

```
<?php
session_start();

if(isset($_SESSION['username'])){
    echo "welcome ".$_SESSION['username'];
}
else{
    $_SESSION['username'] = $_POST['username'];
    $_SESSION['email'] = $_POST['email'];
    echo "session created..";
    exit;
}??>
```

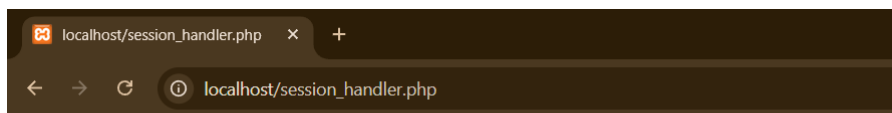
OUTPUT:



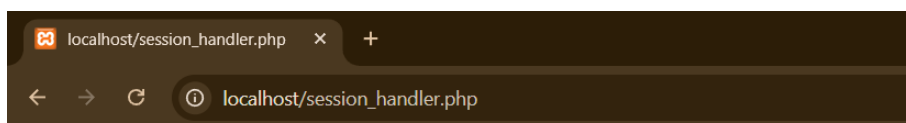
Session Management Form

Name:

Email:



session created..



welcome wxyz

Question: Create a program to write & retrieve cookies

```
<!DOCTYPE html>

<html>

<head>

    <title>Cookie Form</title>

</head>

<body>

    <h1>Cookie Management</h1>

    <form method="POST" action="cookie_set.php">

        <label>Username:</label><br>

        <input type="text" name="username" required><br><br>

        <label>Email:</label><br>

        <input type="email" name="email" required><br><br>

        <button type="submit">Set Cookie</button>

    </form>

    <hr>

    <a href="cookie_view.php">View Cookies</a>

</body>

</html>
```

Cookie_set.php

```
<?php

$username = $_POST['username'];

$email = $_POST['email'];

setcookie("username", $username);
```

```
setcookie("email", $email);
```

```
echo "Cookies set successfully!";
```

```
?>
```

Cookie_view.php

```
<?php
```

```
?>
```

```
<!DOCTYPE html>
```

```
<html>
```

```
<head>
```

```
    <title>View Cookies</title>
```

```
</head>
```

```
<body>
```

```
    <h1>Your Cookies</h1>
```

```
    <?php
```

```
    if (isset($_COOKIE['username'])) {
```

```
        echo "<p>Username: " . $_COOKIE['username'] . "</p>";
```

```
    }
```

```
    if (isset($_COOKIE['email'])) {
```

```
        echo "<p>Email:</strong> " . $_COOKIE['email'] . "</p>";
```

```
    }
```

```
    ?>
```

```
    <hr>
```

```
</body>
```

```
</html>
```


Output:



A screenshot of a web browser with two tabs: 'localhost/session_handler.php' and 'Cookie Form'. The address bar shows 'localhost/cookie_form.html'. The page title is 'Cookie Management'. It contains a form with two input fields: 'Username:' with the value 'abcd' and 'Email:' with the value 'abcd@gmail.com'. Below the fields is a 'Set Cookie' button. A horizontal line separates the form from a link labeled 'View Cookies'.

Cookie Management

Username:
abcd

Email:
abcd@gmail.com

Set Cookie

[View Cookies](#)



Cookies set successfully!



Your Cookies

Username: abcd

Email: abcd@gmail.com

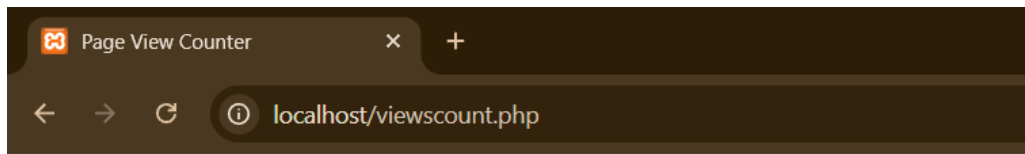
Question: Write a PHP program to store page views count in SESSION, to increment the count oneach refresh, and to show the count on web page.

```
<?php
session_start();
if (isset($_SESSION['views'])) {
    $_SESSION['views'] += 1;
} else {
    $_SESSION['views'] = 1;
}
?>

<!DOCTYPE html>

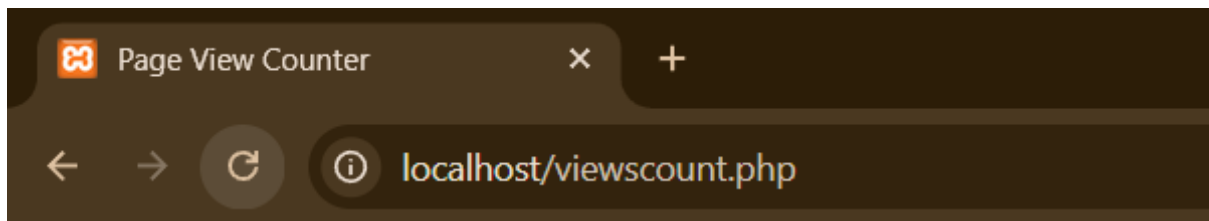
<html>
<head>
    <title>Page View Counter</title>
</head>
<body>
<h2>Page View Counter using Session</h2>
<p>
    You have visited this page
    <strong><?php echo $_SESSION['views']; ?></strong> times in this session.
</p>
</body>
</html>
```

Output:



Page View Counter using Session

You have visited this page **1** times in this session.



Page View Counter using Session

You have visited this page **6** times in this session.

Question: Write a PHP program to store current date-time in a COOKIE and display the "Last visited on date-time on the web page upon reopening of the same page.

```
<?php
$current_time = date('Y-m-d H:i:s');
if (isset($_COOKIE['last_visit'])) {
    $last_visit = $_COOKIE['last_visit'];
    setcookie('last_visit', $current_time, time() + 365*24*60*60);
}
else {
    $last_visit = "First time visiting";
    setcookie('last_visit', $current_time, time() + 365*24*60*60);
}
?>

<!DOCTYPE html>

<html>

<head>

    <title>Last Visit Tracker</title>

</head>

<body>

    <h1>Welcome to the Website</h1>

    <p><strong>Current Time:</strong> <?php echo $current_time; ?></p>

    <p><strong>Last Visited on:</strong> <?php echo $last_visit; ?></p>

    <hr>

</body>

</html>
```

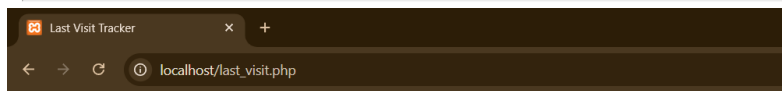
Output:



Welcome to the Website

Current Time: 2026-04-28 14:35:41

Last Visited on: First time visiting



Welcome to the Website

Current Time: 2026-04-28 14:35:53

Last Visited on: 2026-04-28 14:35:41



Welcome to the Website

Current Time: 2026-04-28 14:36:23

Last Visited on: 2026-04-28 14:36:18

Question: Using PHP and MySQL, develop a program to accept book information viz. Accessionnumber, title, authors, edition and publisher from a web page and store the information in a database and to search for a book with the title specified by the user and to display the search results with proper headings.

```
<!DOCTYPE html>

<html>

<head>

    <title>Library Management System</title>

    <style>

        body {

            font-family: Arial;

            margin: 20px;

        }

        .form-section {

            border: 1px solid #ccc;

            padding: 20px;

            margin-bottom: 20px;

            width: 50%;

        }

        input, textarea {

            width: 100%;

            padding: 8px;

            margin: 5px 0;

            box-sizing: border-box;

        }

        button {

            padding: 10px 20px;

            background-color: #4CAF50;

            color: white;

            border: none;

        }

    </style>

</head>

<body>
```

```

        button:hover {
            background-color: #45a049;
        }
    </style>
</head>
<body>
    <h1>Library Management System</h1>
    <div class="form-section">
        <h2>Add Book</h2>
        <form method="POST" action="finder.php">
            <label>Accession Number:</label><br>
            <input type="text" name="accession_number" required><br><br>
            <label>Title:</label><br>
            <input type="text" name="title" required><br><br>
            <label>Author:</label><br>
            <input type="text" name="author" required><br><br>
            <label>Edition:</label><br>
            <input type="text" name="edition" required><br><br>
            <label>Publisher:</label><br>
            <input type="text" name="publisher" required><br><br>
            <button type="submit" name="action" value="Add Book">Add Book</button>
        </form>
    </div>
    <div class="form-section">
        <h2>Search Book</h2>
        <form method="POST" action="finder.php">
            <label>Search by Title:</label><br>
            <input type="text" name="search_title" required><br><br>
            <button type="submit" name="action" value="Search Book">Search</button>
        </form>
    </div>

```

```

        </div>
</body>
</html>

```

Finder.php

```

<?php
$conn = mysqli_connect("localhost", "root", "", "library_db");
if (!$conn) {
    die("Connection failed: " . mysqli_connect_error());
}
$action = isset($_POST['action']) ? $_POST['action'] : "";

if ($action == "Add Book") {
    $a_n = $_POST['accession_number'];
    $title = $_POST['title'];
    $author = $_POST['author'];
    $edition = $_POST['edition'];
    $publisher = $_POST['publisher'];

    $sql = "INSERT INTO books (accession_number, title, authors, edition, publisher)
        VALUES ('$a_n', '$title', '$author', '$edition', '$publisher')";
    if ($conn->query($sql) === TRUE) {
        echo "<h2>Book inserted successfully!</h2>";
    } else {
        echo "<h2>Error: " . $conn->error . "</h2>";
    }
}

else if ($action == "Search Book") {
    $search_title = $_POST['search_title'];
    $q = "SELECT * FROM books WHERE title LIKE '%$search_title%'";
    $result = $conn->query($q);
}

```



```

if ($result->num_rows > 0) {
    echo "<h2>Search Results for: " . $search_title . "</h2>";
    echo "<table border='1' cellpadding='10'>";
    echo "<tr><th>Accession
Number</th><th>Title</th><th>Author</th><th>Edition</th><th>Publisher</th></tr>";
    while ($row = $result->fetch_assoc()) {
        echo "<tr>";
        echo "<td>" . $row['accession_number'] . "</td>";
        echo "<td>" . $row['title'] . "</td>";
        echo "<td>" . $row['authors'] . "</td>";
        echo "<td>" . $row['edition'] . "</td>";
        echo "<td>" . $row['publisher'] . "</td>";
        echo "</tr>";
    }
    echo "</table>";
} else {
    echo "<h2>No books found for: " . $search_title . "</h2>";
}
}

$conn->close();
?>

```

Output



Library Management System

Add Book

Accession Number:

Title:

Author:

Edition:

Publisher:

Search Book

Search by Title:

Library Management System

Add Book

Accession Number:

Title:

Author:

Edition:

Publisher:

Search Book

Search by Title:

Search Results for: The Atomic Habits

Accession Number	Title	Author	Edition	Publisher
123	The Atomic Habits	James clear	2022	Om

[Back to Form](#)

Question: Create a login form with fields for User ID and Password. Upon form submission, validate the entered credentials by matching them with existing records. If the User ID and Password are correct, display a new welcome page.

```
<!DOCTYPE html>

<html>

<head>

  <title>Login Form</title>

  <style>

    body {

      font-family: Arial;

      margin: 20px;

    }

    input {

      padding: 8px;

      margin: 10px 0;

      width: 200px;

    }

    button {

      padding: 10px 20px;

      background-color: #4CAF50;

      color: white;

      border: none;

      cursor: pointer;

    }

  </style>

</head>

<body>

  <h1>User Login</h1>

  <form method="POST" action="login_process.php">

    <label>User ID:</label><br>

    <input type="text" name="user_id" required><br>
```

```

        <label>Password:</label><br>
        <input type="password" name="password" required><br>
        <button type="submit">Login</button>
    </form>
</body>
</html>

```

Login_process.php

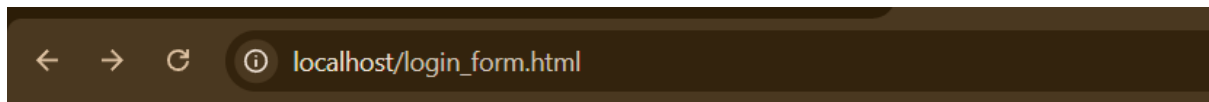
```

<?php
session_start();
$conn = mysqli_connect("localhost", "root", "", "library_db");
if (!$conn)
    die("Connection failed: " . mysqli_connect_error());
if ($_SERVER['REQUEST_METHOD'] === 'POST') {
    $user_id = $_POST['user_id'];
    $password = $_POST['password'];
    $q = "SELECT * FROM users WHERE user_id = '$user_id' AND password = '$password'";
    $result = $conn->query($q);
    if ($result->num_rows > 0) {
        $row = $result->fetch_assoc();
        $_SESSION['user_id'] = $row['user_id'];
        header("Location: welcome.php");
        exit;
    } else {
        echo "<h1>Login Failed</h1>";
        echo "<p>Invalid User ID or Password</p>";
        echo "<a href='login_form.html'>Try Again</a>";
    }
}
$conn->close();
?>

```

Output:

← T →				user_id	password
☐	 Edit	 Copy	 Delete	admin	admin123
☐	 Edit	 Copy	 Delete	user	pass123



User Login

User ID:

Password:

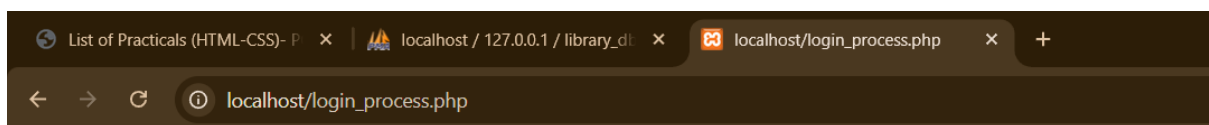


Welcome admin

You have successfully logged in!

Your User ID: admin

[Logout](#)



Login Failed

Invalid User ID or Password

[Try Again](#)

Question: Create a PHP page that displays all records from the PERS table where the Department Number (dno) matches one of the values listed on the web interface.

```
<!DOCTYPE html>

<html>

<head>

  <title>PERS Records by Department</title>

  <style>

    body {

      font-family: Arial;

      margin: 20px;

    }

    .form-container {

      border: 1px solid #ccc;

      padding: 20px;

      width: 50%;

    }

    input, select {

      padding: 8px;

      margin: 10px 0;

      width: 100%;

      box-sizing: border-box;

    }

    button {

      padding: 10px 20px;

      background-color: #4CAF50;

      color: white;

      border: none;

      cursor: pointer;

      margin-top: 10px;

    }

  </style>
```

```

</head>
<body>
    <h1>PERS Records by Department</h1>
    <div class="form-container">
        <form method="POST" action="pers_display.php">
            <label>Select Department Number(s):</label><br>
            <select name="dno[]" multiple required>
                <option value="1">Department 1</option>
                <option value="2">Department 2</option>
                <option value="3">Department 3</option>
                <option value="4">Department 4</option>
                <option value="5">Department 5</option>
            </select>
            <small>Hold Ctrl (or Cmd on Mac) to select multiple departments</small><br><br>
            <button type="submit">Display Records</button>
        </form>
    </div>
</body>
</html>

```

Pers_display.php

```

<?php
$conn = mysqli_connect("localhost", "root", "", "library_db");
if (!$conn) {
    die("Connection failed: " . mysqli_connect_error());
}
$dno_array = $_POST['dno'];
$dno_list = implode(',', $dno_array);
$q = "SELECT * FROM pers WHERE dno IN ($dno_list)";
$result = $conn->query($q);
if ($result->num_rows > 0) {

```

```

echo "<h2>PERS Records for Selected Departments</h2>";
echo "<table border='1' cellpadding='10'>";
echo "<tr>";
$fields = $result->fetch_fields();
foreach ($fields as $field) {
    echo "<th>" . $field->name . "</th>";
}
echo "</tr>";
while ($row = $result->fetch_assoc()) {
    echo "<tr>";
    foreach ($row as $value) {
        echo "<td>" . $value . "</td>";
    }
    echo "</tr>";
}

echo "</table>";
}
else {
    echo "<h2>No records found for selected departments</h2>";
}
?>

<br>
<a href="pers_form.html">Back to Form</a>

```


Output:

☐ Show all | Number of rows: 25 | Filter rows: Search this table | Sc

Extra options

				eid	ename	esalary	dno	designation
☐	Edit	Copy	Delete	101	John Smith	50000.00	1	Manager
☐	Edit	Copy	Delete	102	Sarah Johnson	45000.00	1	Developer
☐	Edit	Copy	Delete	103	Mike Brown	55000.00	2	Senior Developer
☐	Edit	Copy	Delete	104	Emma Davis	48000.00	2	Analyst
☐	Edit	Copy	Delete	105	Robert Wilson	52000.00	3	Manager
☐	Edit	Copy	Delete	106	Lisa Anderson	47000.00	3	Coordinator
☐	Edit	Copy	Delete	107	James Taylor	51000.00	4	Engineer
☐	Edit	Copy	Delete	108	Mary Miller	46000.00	4	Technician
☐	Edit	Copy	Delete	109	David Martinez	53000.00	5	Supervisor
☐	Edit	Copy	Delete	110	Jennifer Garcia	49000.00	5	Associate

☐ Check all | With selected: Edit Copy Delete Export

[←](#) [→](#) [↻](#) [ⓘ](#) localhost/pers_form.html

PERS Records by Department

Select Department Number(s):

Department 1

Department 2

Department 3

Department 4

Department 5

Hold Ctrl (or Cmd on Mac) to select multiple departments

Display Records

[←](#) [→](#) [↻](#) [ⓘ](#) localhost/pers_display.php

PERS Records for Selected Departments

eid	ename	esalary	dno	designation
103	Mike Brown	55000.00	2	Senior Developer
104	Emma Davis	48000.00	2	Analyst

[Back to Form](#)

REACT

Q 1: Create a functional component that accepts props and displays a personalized message.

App.js

```
import logo from './logo.svg';
import './App.css';
import Cmp2 from './Cmp2';
function App() {
  return (
    <div className="App">
      <Cmp2 name="Gehu123" age="21"/>
    </div>
  );
}
export default App;
```

Cmp2.jsx

```
function Cmp2(props) {
  return (
    <div>
      <h2>Hello {props.name}, your age is {props.age}</h2>
    </div>
  );
}
export default Cmp2;
```

OUTPUT:



Hello Gehu123, your age is 21

Q 2: Create a functional component that accepts props and displays a personalized message.

App.js

```
import logo from './logo.svg';
import './App.css';
import Counter1 from './Counter1';

function App() {
  return (
    <div className="App">
      <Counter1/>
    </div>
  );
}
export default App;
```

Counter1.jsx

```
import React, { Component } from "react";

class Counter1 extends Component {
  state = { count: 0 };
  increase = () => {
    this.setState(prev => ({ count: prev.count + 1 }));
  };
  decrease = () => {
    this.setState(prev => ({ count: prev.count - 1 }));
  };
  update() {
    return (
      <div> <h2>Count: {this.state.count}</h2>
      <button onClick={this.increase}>Increase</button>
      <button onClick={this.decrease}>Decrease</button>
    </div>
    );
  }
}
export default Counter1;
```

OUTPUT:



Q 3: Create a functional component that maintains a counter and provides buttons to increase, decrease and reset the counter.

App.js

```
import logo from './logo.svg';
import './App.css';
import Counter from './Counter';
```

```
function App() {
  return (
    <div className="App">
      <Counter/>
    </div>
  );
}
export default App;
```

Counter.jsx

```
import { useState } from "react";
function Counter() {
  const [count, setCount] = useState(0);
  return (
    <div style={{ textAlign: "center", marginTop: "50px" }}>
      <h2>Counter: {count}</h2>
      <button onClick={() => setCount(count + 1)}>Increase</button>
      <button onClick={() => setCount(count - 1)} style={{ marginLeft: "10px" }}>
        Decrease </button>
      <button onClick={() => setCount(0)} style={{ marginLeft: "10px" }}>
        Reset </button>
    </div>
  );
}
export default Counter;
```

OUTPUT:



Q 4: Create a functional component that displays a running clock and the current date.

App.js

```
import logo from './logo.svg';
import './App.css';
import Clock from './Clock';
```

```
function App() {
  return (
    <div className="App">
      <Clock/>
    </div>
  );
}
export default App;
```

Clock.jsx

```
import { useState, useEffect } from "react";
function Clock() {
  const [time, setTime] = useState(new Date());

  useEffect(() => {
    const timer = setInterval(() => {
      setTime(new Date());
    }, 1000);
    return () => clearInterval(timer);
  }, []);

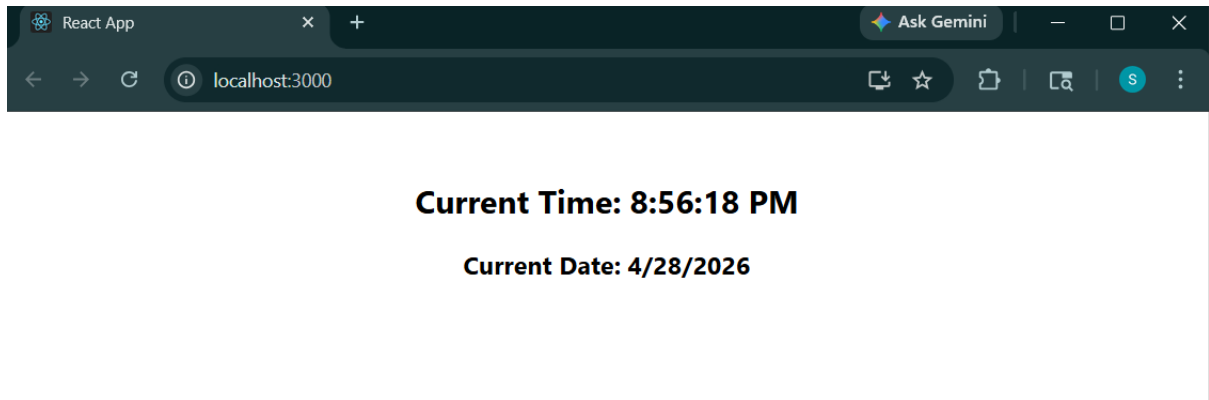
  return (
    <div style={{ textAlign: "center", marginTop: "50px" }}>
      <h2>Current Time: {time.toLocaleTimeString()}</h2>
    </div>
  );
}
```

```
    <h3>Current Date: {time.toLocaleDateString()}</h3>
  </div>

);
}

export default Clock;
```

OUTPUT:



Q 5: Create a component with a form that updates the state based on user input.

App.js

```
import logo from './logo.svg';
import './App.css';
import Form from './Form';
function App() {
  return (
    <div className="App">
      <Form/>
    </div>
  );
}
export default App;
```

Form.jsx

```
import { useState } from "react";
function UserForm() {
  const [formData, setFormData] = useState({
    name: "",
    email: "",
    age: ""
  });
  function handleChange(e) {
    const { name, value } = e.target;
    setFormData(prev => ({
      ...prev,
      [name]: value
    }));
  }
  function handleSubmit(e) {
```

```

    e.preventDefault();

    alert(`Name: ${formData.name}\nEmail: ${formData.email}\nAge: ${formData.age}`);
  }

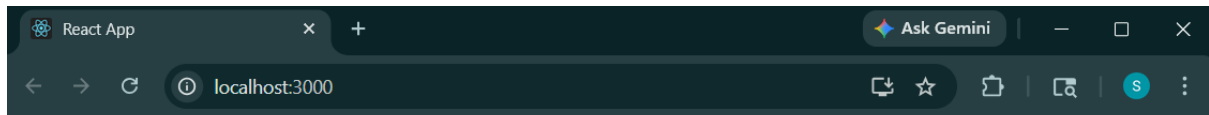
  return (
    <div style={{ textAlign: "center", marginTop: "40px" }}>
      <h2>Registration Form</h2>
      <form onSubmit={handleSubmit}>
        <input type="text" name="name" placeholder="Enter Name"
          value={formData.name} onChange={handleChange}/>
        <br /><br />
        <input type="email" name="email" placeholder="Enter Email"
          value={formData.email} onChange={handleChange}/>
        <br /><br />

        <input
          type="number" name="age" placeholder="Enter Age" value={formData.age}
          onChange={handleChange} />
        <br /><br />
        <button type="submit">Submit</button>
      </form>
      <h3>Live Preview</h3>
      <p>Name: {formData.name}</p>
      <p>Email: {formData.email}</p>
      <p>Age: {formData.age}</p>
    </div>
  );
}

export default UserForm;

```

OUTPUT:



Registration Form

Live Preview

Name:

Email:

Age:

React App

Ask Gemini

localhost:3000

Registration Form

Live Preview

Name: David Warner

Email: david@gmail.com

Age: 37

Q 6: Create a React application with multiple routes.

App.js

```
import logo from './logo.svg';
import './App.css';
import Route from './Route';
function App() {
  return (
    <div className="App">
      <Route/>
    </div>
  );
}
export default App;
```

Route.jsx

```
import { BrowserRouter as Router, Routes, Route, Link } from "react-router-dom";
function Home() {
  return <h2>Welcome to Home Page</h2>;
}
function About() {
  return <h2>This is About Page</h2>;
}
function Contact() {
  return <h2>This is Contact Page</h2>;
}
function App() {
  return (
    <Router>
      <div style={{ textAlign: "center", marginTop: "40px" }}>
        <h1>React Routing Example</h1>

```

```

<nav>

  <Link to="/">Home</Link>{" | "}

  <Link to="/about">About</Link>{" | "}

  <Link to="/contact">Contact</Link>

</nav> <br /><br />

<Routes>

  <Route path="/" element={<Home />} />

  <Route path="/about" element={<About />} />

  <Route path="/contact" element={<Contact />} />

</Routes>

</div>

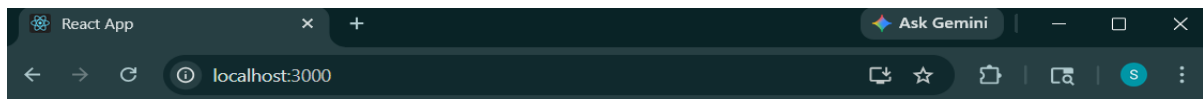
</Router>

);
}

export default App;

```

OUTPUT:



React Routing Example

[Home](#) | [About](#) | [Contact](#)

Welcome to Home Page



React Routing Example

[Home](#) | [About](#) | [Contact](#)

This is About Page

React Routing Example

[Home](#) | [About](#) | [Contact](#)

This is Contact Page